K230 Vision Obstacle Avoidance Car - Project Report

1. Overview

This project uses a K230 / CanMV vision module to detect red obstacles and assist a smart car in avoiding

2. Vision Pipeline

- Capture 320 x 180 RGB565 frames from the camera.
- Detect red image regions using LAB thresholds.
- Merge red cap/label regions and infer complete bottle-like obstacle boxes.
- Build an image-space driving corridor from fixed or calibrated black floor-tape lines.
- Treat an obstacle as blocking when its bottom-center point lies inside the corridor.

3. UART Protocol

TI -> K230: 12 FF starts stage 1, 12 1B enables vision feedback, 13 FF stops feedback for stage 2.

K230 -> TI: 12 00 means blocked, 12 01 means corridor clear.

4. Tuning Notes

Tune RED_THRESHOLDS and BLACK_THRESHOLDS on the real field. Use RAW, BOTTLE, CORRIDOR, and LI

5. Repository Notes

Replace images/car_demo.jpg with a real car photo and videos/demo_link.txt with a public demo video link